

SHIV NADAR

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DELHI NCR

Prompt-Guided Object Material Editing using Intrinsic Decomposition, Segmentation, and Diffusion Refinement

Course: Computer Vision CSD454

Professor: Sumit Shekhar

Team members:

NAMIT BANSAL – 2310110531

AYUSH TIWARI – 2310110661

PARTH AGARWAL – 2310110546

KRISHNA GARG – 2310110692

Department of Computer Science and Engineering

Shiv Nadar University

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ABSTRACT

Editing the material appearance of objects in images while preserving geometry remains a challenging problem in computer vision. Recent generative models, particularly diffusion models, can produce highly realistic images but struggle to maintain structural consistency during targeted image editing tasks. This project investigates the suitability of diffusion models for material editing and proposes a hybrid pipeline combining object segmentation, intrinsic image decomposition, frequency-domain material manipulation, and optional prompt-guided LoRA diffusion refinement.

During development, diffusion models—even when fine-tuned with domain-specific LoRA adapters—proved fundamentally misaligned with the material editing task. This misalignment arises because diffusion models operate in pixel or latent space where geometry, lighting, and material properties are entangled.

To address these challenges, the final system employs a physics-inspired pipeline in which intrinsic editing performs the primary material transformation and diffusion contributes only limited visual refinement. The system includes an interactive Streamlit interface allowing users to upload images, automatically segment foreground objects, apply material transformations through a continuous slider, and analyze the impact of diffusion refinement.

Experimental results show that the physics-based editing pipeline produces geometry-preserving material transformations more reliably than diffusion-only approaches. The project demonstrates that integrating physically meaningful representations with modern generative models provides a promising direction for structured image editing tasks.

Keywords: Material Editing, Intrinsic Image Decomposition, Diffusion Models, Object Segmentation, Reflectance Modeling, Frequency Analysis, LoRA Fine-Tuning.

1 INTRODUCTION

Image editing technologies play a critical role in modern digital workflows across fields such as product visualization, augmented reality, film production, game design, and e-commerce. Among the various image manipulation tasks, modifying the material properties of objects presents unique challenges.

Changing the material of an object—for example transforming a plastic object into a metallic one—requires altering reflectance characteristics while preserving the object’s geometry, structure, and lighting context. Human perception of materials is highly sensitive to subtle cues such as specular highlights, surface reflections, and shading gradients.

Traditional image editing tools typically rely on manual adjustments to colour, brightness, and contrast. While these techniques can simulate superficial appearance changes, they do not accurately reproduce the complex reflectance properties associated with real materials.

Recent advances in generative models have introduced powerful tools capable of producing highly realistic images. Diffusion models, in particular, have demonstrated remarkable success in text-to-image generation tasks. These models learn a probabilistic representation of natural images and generate samples through iterative denoising processes.

However, despite their impressive generative capabilities, diffusion models face significant limitations when applied to controlled editing tasks. When used for image editing,

diffusion models often alter object structure, distort boundaries, or modify unrelated regions of the image.

The central hypothesis of this project was that a diffusion model fine-tuned using Low-Rank Adaptation (LoRA) could serve as a material editor when guided by appropriate prompts. Systematic experimentation revealed that diffusion models struggle to reliably perform such transformations while preserving object geometry.

These observations motivated the development of a hybrid editing pipeline in which physically grounded image representations guide the editing process, while diffusion contributes only minor perceptual refinements.

The final system integrates:

- object segmentation
- intrinsic image decomposition
- frequency-domain material editing
- optional diffusion-based refinement

This combination allows the system to modify material properties while preserving structural consistency.

2 RELATED WORK

2.1 Intrinsic Image Decomposition

Intrinsic image decomposition aims to separate an image into reflectance and illumination components. This decomposition provides a physically meaningful representation of the scene because material properties primarily affect reflectance rather than illumination.

Barron and Malik proposed methods for estimating reflectance and illumination from a single image using statistical priors. Bell et al. introduced the Intrinsic Images in the Wild dataset, which enabled data-driven approaches to intrinsic decomposition.

Intrinsic decomposition is particularly useful for image editing because it allows modifications to be applied to reflectance components without affecting illumination structure.

2.2 Diffusion Models

Diffusion models generate images through a stochastic process that gradually removes noise from an initial random signal. The training process teaches the model to reverse a noise-corruption process applied to images.

Ho et al. introduced Denoising Diffusion Probabilistic Models, which demonstrated that iterative denoising could produce high-quality images. Later work by Rombach et al. introduced latent diffusion models, which perform diffusion in a compressed latent space to improve computational efficiency.

These models form the foundation of widely used systems such as Stable Diffusion.

2.3 Diffusion-Based Image Editing

Several approaches have attempted to adapt diffusion models for editing tasks. Prompt-to-Prompt editing modifies cross-attention maps to guide semantic transformations. DiffEdit introduces masked diffusion editing to localize modifications to specific regions of the image.

Despite these advances, diffusion-based editing methods often struggle with structural preservation because the underlying generative objective encourages the model to produce plausible images rather than maintain strict correspondence with the input image.

2.4 Object Segmentation

Accurate object segmentation is essential for localized image editing. The Segment Anything Model (SAM) provides a general-purpose segmentation framework trained on large-scale datasets. SAM can generate segmentation masks for a wide variety of objects without requiring task-specific training.

In this project, segmentation is used to ensure that material edits affect only the object of interest while preserving the background.

2.5 Research Gaps and Motivation

Although significant progress has been made in image generation and editing, several important limitations remain when attempting to perform controlled material editing.

One major challenge arises from the entanglement of geometry and appearance in generative models. Diffusion models learn correlations between object shape, lighting conditions, and material properties from training data. As a result, attempting to change material appearance often leads to unintended changes in object structure.

Another limitation is the absence of physically interpretable representations in generative models. Material properties are governed by reflectance characteristics that interact with illumination and geometry. Without explicitly modelling these components, it becomes difficult to apply controlled material transformations.

Spatial localization also remains challenging. Many editing methods operate on entire images rather than focusing on specific objects, leading to unintended changes in background regions.

Finally, prompt-based editing provides only coarse control over visual attributes. Text prompts cannot easily specify detailed material properties such as specular intensity, surface roughness, or micro-texture patterns.

These limitations motivate the need for a hybrid system that combines physically meaningful image representations with generative refinement techniques.

3 Dataset Preparation

3.1 Base Dataset Selection

The Extended Complex Scene Saliency Dataset (ECSSD) was selected as the base dataset for training and experimentation. ECSSD contains images of natural scenes with corresponding saliency masks that highlight prominent objects.

The dataset was chosen because it provides images where the foreground object is clearly distinguishable from the background, making it suitable for object-focused editing tasks.

A subset of 300 images was selected based on:

- object prominence
- mask quality
- visual diversity



Figure 1: Example Images from ECSSD Dataset

3.2 Intrinsic Decomposition of Training Images

Each selected image was decomposed into intrinsic components representing:

- albedo
- shading
- specularity

These components capture different aspects of the physical image formation process. The decomposition process produces separate images representing reflectance and illumination characteristics.

3.3 Synthetic Material Edit Generation

A parametric editing pipeline was developed to generate synthetic material variations from the intrinsic components. Edits were restricted to the object region defined by the segmentation mask.

Two primary material categories were simulated:

- metallic surfaces
- plastic surfaces

Material variations were produced by sampling parameters controlling reflectance and shading properties.

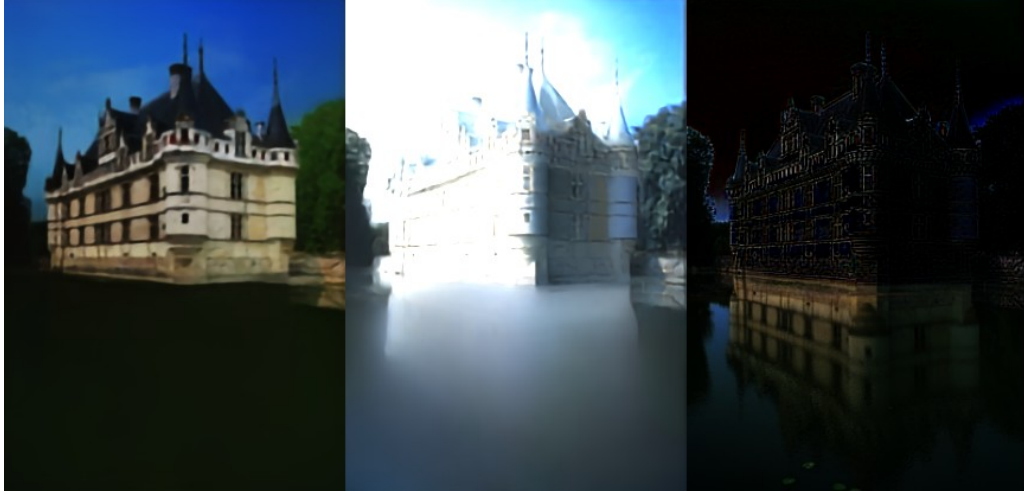


Figure 2: Intrinsic Decomposition Example

Category	Count
Base images	300
Metal edits	~30,000
Plastic edits	~30,000
Total samples	~60,000

Table 1: Dataset Summary

4 Initial Approach: LoRA Diffusion Model

The project initially explored the use of diffusion models as the primary material editing mechanism. Diffusion models have demonstrated exceptional performance in text-guided image synthesis and style transfer tasks. The hypothesis was that a diffusion model fine-tuned with domain-specific training data could perform material editing while preserving object geometry.

To test this hypothesis, a LoRA (Low-Rank Adaptation) fine-tuning strategy was used to adapt the Stable Diffusion model to the material editing task.

LoRA was chosen because it allows efficient adaptation of large models with minimal additional parameters. Instead of updating all model weights, LoRA introduces low-rank matrices that modify the attention layers of the diffusion model.

4.1 Diffusion Model Architecture

The base model used for training was Stable Diffusion 1.5, which consists of three primary components:

- Variational Autoencoder (VAE)
- UNet denoising network
- Text encoder (CLIP)

The training objective of diffusion models is to learn the reverse of a noise corruption process.



Figure 3: Synthetic Material Editing Examples

During training, an image is progressively corrupted with Gaussian noise:

$$x_t = \alpha_t x_0 + (1 - \alpha_t) \epsilon$$

where:

x_t represents the noisy image at timestep t

ϵ is random Gaussian noise.

The model learns to predict the noise component ϵ , enabling it to reconstruct the original image through iterative denoising.

4.2 LoRA Fine-Tuning Strategy

Instead of retraining the full diffusion model, LoRA adapters were applied to the attention layers of the UNet network. The weight update can be represented as:

$$W' = W + \Delta W$$

where

$$\Delta W = AB$$

Here A and B are low-rank matrices. This significantly reduces the number of trainable parameters while allowing the model to adapt to new tasks.

4.3 Training Configuration

The LoRA model was trained using the synthetic dataset described earlier.

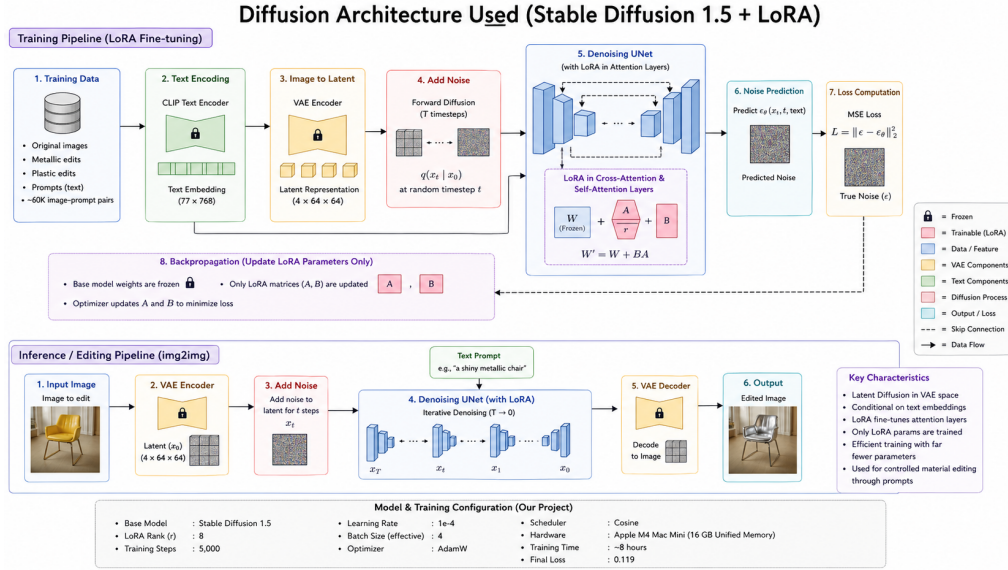


Figure 4: Diffusion Model Architecture

Parameter	Value
Base model	Stable Diffusion 1.5
LoRA rank	8
Learning rate	1×10^{-4}
Training steps	5000
Effective batch size	4
Scheduler	Cosine
Hardware	Apple Silicon M4
Training time	~8 hours
Final loss	0.119

Table 2: Training Configuration

4.4 Observed Limitations

During experimentation, two critical behaviours were observed:

1. Low img2img strength

The model produced minimal visible material change.

2. High img2img strength

The model generated stronger material textures but introduced:

- object boundary distortion
- geometric changes
- background artifacts.

This indicated that diffusion models struggle to balance structure preservation and appearance transformation.

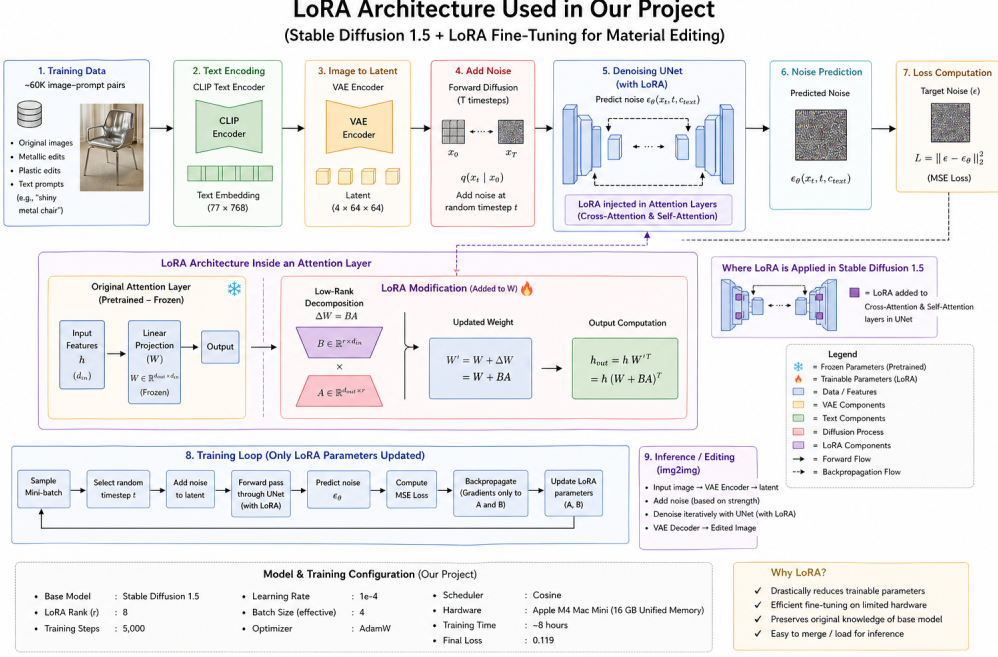


Figure 5: LoRA Training Pipeline

5 Why Diffusion Models Fail at Material Editing

The observed limitations are not merely due to hyperparameters but arise from the fundamental design of diffusion models.

The physical process of image formation can be approximated as:

$$I = A \times S + R$$

where:

A = Albedo

S = Shading

R = Specular reflection

Material changes affect primarily the reflectance and specular components rather than geometry. However, diffusion models operate in latent feature space, where these components are entangled.

5.1 Lack of Explicit Material Representation

Diffusion models do not explicitly represent:

- reflectance
- illumination
- specular reflections.

Instead, these properties are implicitly encoded in learned latent features. This makes it difficult to isolate material changes without affecting other visual attributes.

5.2 Synthetic Dataset Limitations

The training dataset consisted of synthetic material transformations generated from intrinsic components. Although useful, this dataset has limited diversity compared to real-world material variations. Diffusion models typically require millions of training samples to capture subtle appearance changes.

5.3 High-Frequency Material Signals

Human perception of materials depends heavily on high-frequency signals such as:

- micro-scratches
- highlight gradients
- reflection sharpness.

These features are difficult to learn using small LoRA adapters.

5.4 Strength–Structure Trade-off

In diffusion editing, the strength parameter controls how much the output deviates from the input.

Low strength:

- preserves structure
- weak material change

High strength:

- strong appearance change
- geometry distortion.

This trade-off makes diffusion unsuitable as the primary editing mechanism.

6 Physics-Based Material Editing Pipeline

Due to the limitations observed with diffusion models, a physics-inspired editing pipeline was developed. The pipeline focuses on direct manipulation of intrinsic image components.

6.1 Automatic Object Segmentation

Accurate object segmentation is required to ensure that material edits affect only the object. Instead of using heavy segmentation models, a lightweight method combining saliency detection and GrabCut refinement was implemented.

The process includes:

1. Frequency-based saliency detection

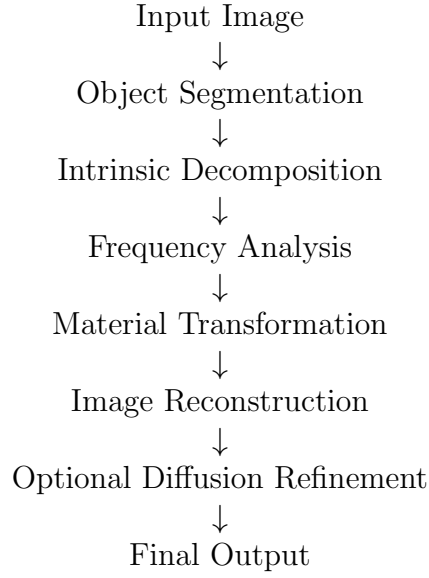


Figure 6: Proposed System Pipeline

2. Foreground bounding box estimation
3. GrabCut graph-based segmentation
4. Morphological cleanup
5. Gaussian smoothing of mask edges.



Figure 7: Segmentation Process

6.2 Intrinsic Image Decomposition

The image is decomposed into three components:

- albedo
- shading
- specular.

A fast approximation suitable for real-time interaction is used. The albedo component is estimated using:

$$\text{Albedo} = \frac{\text{Image}}{\text{Shading} + \epsilon}$$

where ϵ prevents numerical instability.



Figure 8: Frequency Decomposition Components

6.3 Frequency Band Extraction

Material appearance strongly correlates with frequency patterns. Metal surfaces contain:

- high-frequency reflections
- sharp highlights.

Plastic surfaces contain:

- smoother reflections
- softer gradients.

The pipeline extracts:

- mid-frequency shading bands
- high-frequency specular bands.

These bands are manipulated to simulate different materials.

6.4 Metallic Material Transformation

Metallic materials exhibit:

- strong specular reflections
- reduced colour saturation

- high-frequency surface scratches.

The editing process modifies the intrinsic components accordingly.

Albedo modification:

$$A' = \text{desaturate}(A) \times \text{darken}(A)$$

Specular amplification:

$$Sp' = Sp + \alpha \cdot \text{high_frequency}$$

Shading sharpening:

$$Sd' = Sd - \beta \cdot \text{mid_frequency}$$

Metal-specific colour tints such as silver, gold, and copper are applied.

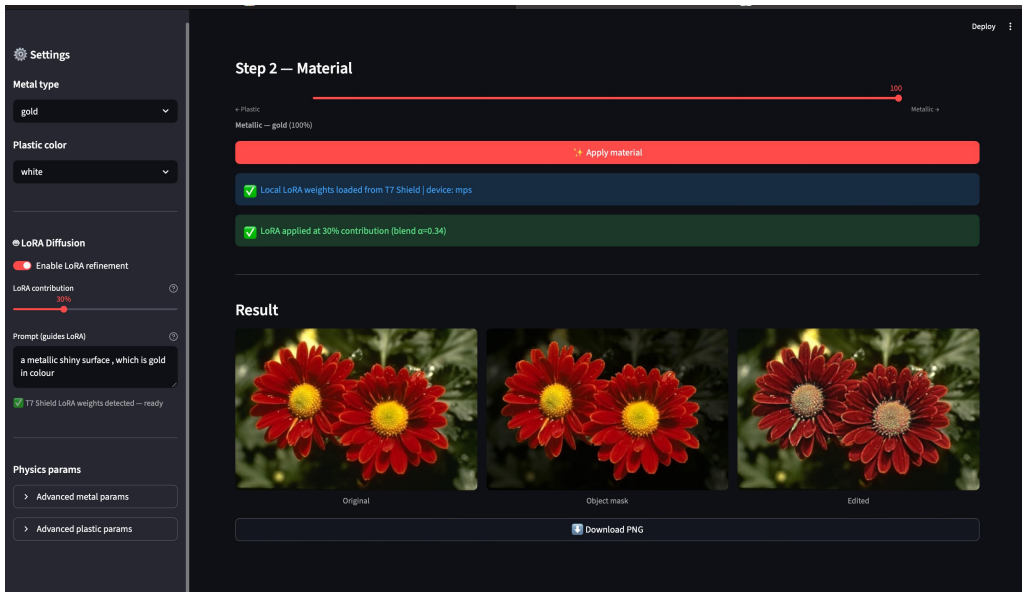


Figure 9: Metallic Transformation Example

6.5 Plastic Material Transformation

Plastic materials exhibit different visual characteristics. Typical properties include:

- smooth reflections
- vivid colour
- soft highlights.

The plastic editing pipeline performs:

- colour saturation boost
- smooth shading preservation
- broad highlight generation.

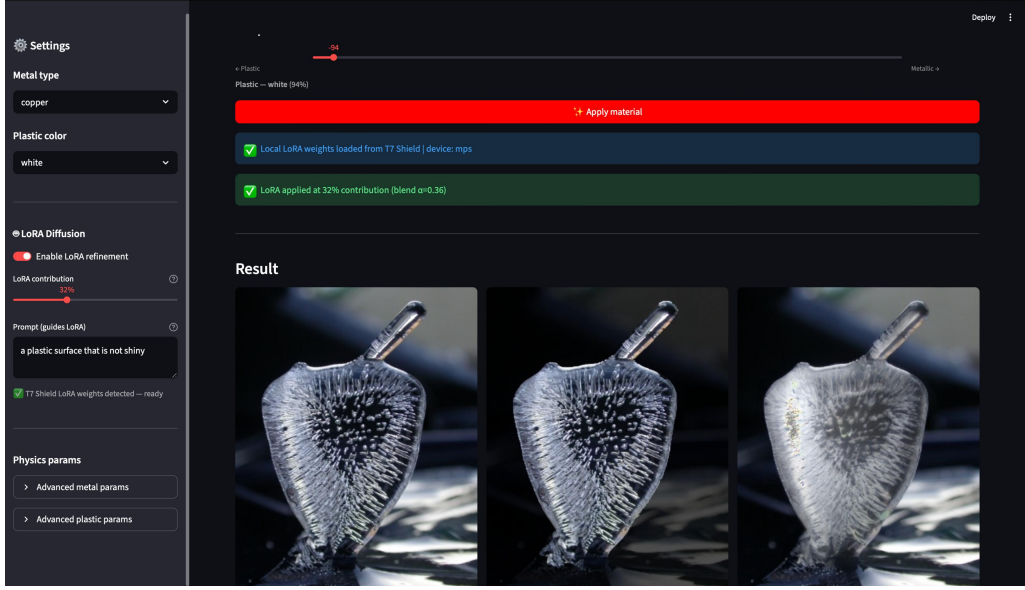


Figure 10: Plastic Transformation Example

6.6 Image Reconstruction

After modifying intrinsic components, the final image is reconstructed.

$$\text{Edited} = A' \times Sd' + Sp'$$

To preserve the background:

$$\text{Final} = \text{mask} \cdot \text{Edited} + (1 - \text{mask}) \cdot \text{Original}$$

This ensures that only the object region is modified.

7 System Architecture and Implementation

The final system integrates segmentation, intrinsic image decomposition, frequency-domain editing, and optional diffusion refinement into a single processing pipeline. The architecture is designed to separate physically grounded image manipulation from generative refinement, ensuring that structural changes are minimized.

The system is implemented primarily using Python with the following libraries:

Component	Implementation
Image processing	NumPy, OpenCV
Segmentation	GrabCut + saliency detection
Intrinsic decomposition	Custom Gaussian-based approximation
Material editing	Frequency-domain manipulation
Diffusion refinement	Stable Diffusion 1.5 via HuggingFace Diffusers
Training framework	HuggingFace Accelerate
Hardware	Apple Silicon M4 Mac Mini

The pipeline processes an image through the following stages:

1. Image loading and resizing
2. Object segmentation
3. Intrinsic image decomposition
4. Frequency band extraction
5. Material transformation
6. Image reconstruction
7. Optional diffusion refinement

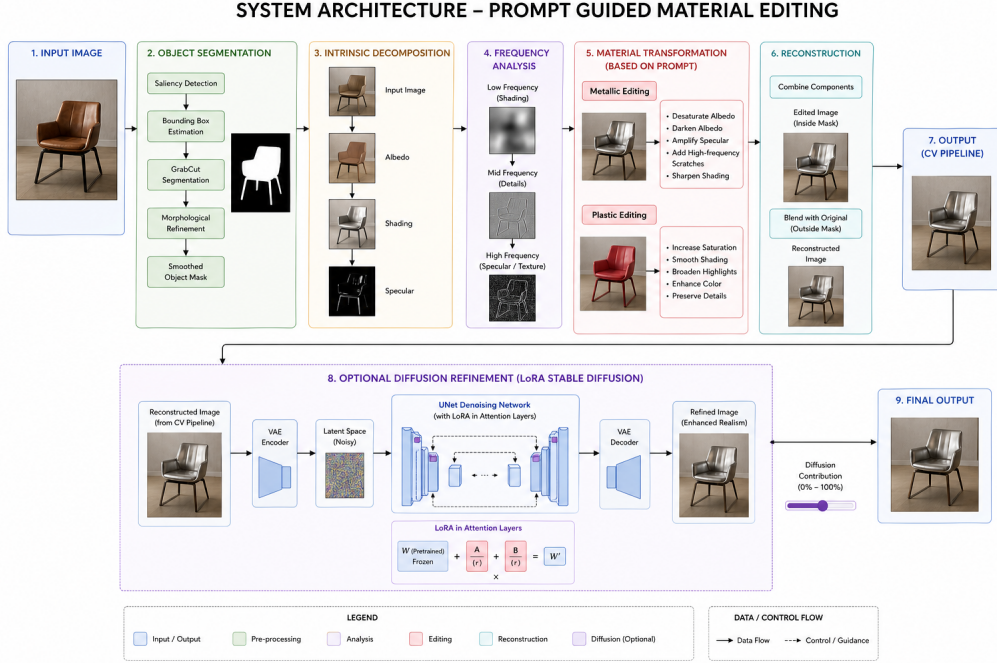


Figure 11: Full System Architecture

8 Experimental Setup

8.1 Dataset

The ECSSD dataset was used as the base dataset due to its availability of high-quality object masks and diverse natural scenes. Dataset preparation involved selecting a subset of 300 images with clearly defined foreground objects.

Each selected image underwent the following preprocessing steps:

- segmentation mask generation
- intrinsic decomposition
- frequency band extraction

Synthetic material edits were generated through parameterized transformations applied to the intrinsic components.

The resulting dataset contained approximately:

Dataset Category	Image Count
Base images	300
Metal edits	~30,000
Plastic edits	~30,000
Total training samples	~60,000

8.2 Training Procedure

The diffusion refinement model was trained using LoRA fine-tuning on Stable Diffusion 1.5.

Training configuration:

Parameter	Value
Base model	runwayml/stable-diffusion-v1-5
LoRA rank	8
Training steps	5000
Effective batch size	4
Learning rate	1×10^{-4}
Scheduler	Cosine
Training time	~8 hours
Final loss	0.119

Training was performed on an Apple Silicon system using the MPS backend.

9 Experimental Results

The system was evaluated qualitatively by examining the visual realism and structural consistency of edited images. The primary goal was to verify whether the system could modify material appearance while preserving object geometry and background context.

9.1 Metallic Material Edits

Metallic transformations exhibited the following characteristics:

- reduced color saturation
- enhanced specular reflections
- sharper highlight gradients
- metallic tint variations (silver, gold, copper)

These properties are consistent with known physical characteristics of metallic surfaces.

9.2 Plastic Material Edits

Plastic transformations demonstrated:

- increased colour saturation
- smoother specular highlights
- softer shading transitions
- vivid surface colours

These characteristics correspond to common visual properties of plastic materials.

9.3 Geometry Preservation

One of the primary objectives of the project was maintaining object geometry during editing. The physics-based editing pipeline achieved consistent geometry preservation because transformations were applied only to intrinsic components rather than pixel-level generative modifications. Background regions remained unchanged due to segmentation-based masking.

9.4 Diffusion Contribution Analysis

To understand the role of diffusion in the system, the LoRA diffusion model was incorporated as an optional refinement stage. The diffusion contribution slider allowed systematic exploration of diffusion influence.

Diffusion Contribution	Observed Behaviour
0–20%	minimal changes, mostly CV editing
20–50%	slight texture refinement
50–80%	stronger material cues but minor distortions
80–100%	significant geometry distortion

These results indicate that diffusion models are better suited for texture enhancement rather than structural editing.

9.5 Runtime Performance

Runtime performance was measured for different stages of the pipeline.

Operation	Approximate Time
Auto segmentation	1–3 seconds
Intrinsic decomposition	< 0.5 seconds
Material editing	< 0.5 seconds
Diffusion refinement	25–40 seconds

The physics-based editing pipeline runs significantly faster than diffusion refinement.

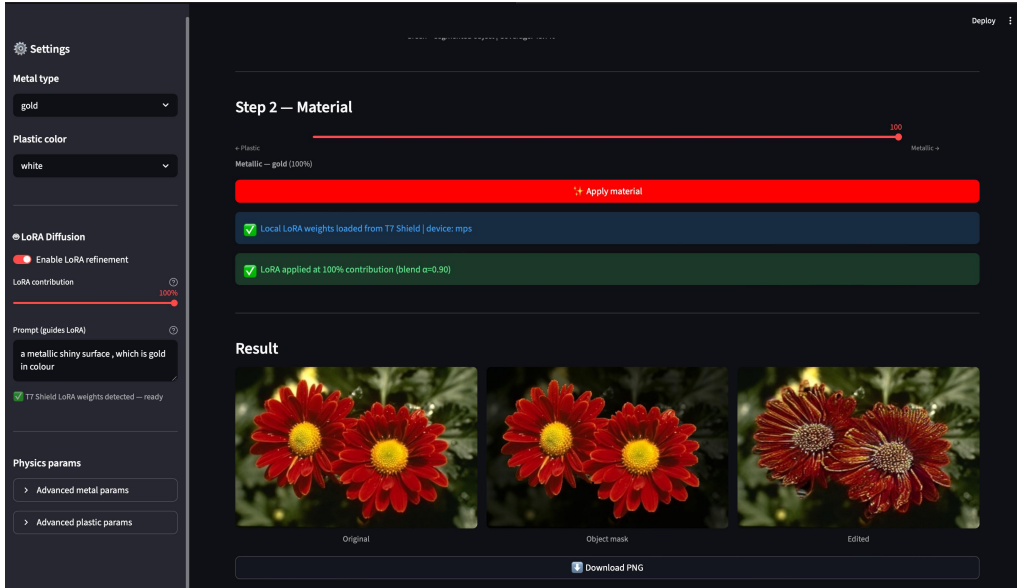


Figure 12: High Diffusion Contribution

10 Discussion

The experiments highlight an important distinction between generative models and structured editing systems. Diffusion models optimize a generative objective that encourages the production of visually plausible images. However, this objective does not enforce strict correspondence with the input image.

In contrast, the physics-based pipeline operates directly on intrinsic components of the image, enabling controlled manipulation of material properties. This approach ensures that geometry and scene structure remain unchanged.

The results suggest that diffusion models are better used as secondary refinement mechanisms rather than primary editing tools.

11 Limitations

Despite its advantages, the proposed system has several limitations. First, the intrinsic decomposition method used in this project is an approximation and may not accurately separate reflectance and illumination in complex scenes. Second, the system cannot fully simulate complex environment reflections or anisotropic material properties. Third, segmentation errors can occasionally lead to artifacts near object boundaries. Finally, the diffusion refinement stage is limited by the capacity of the LoRA model and the relatively small training dataset.

12 Future Work

Several improvements could enhance the system:

- training diffusion models on larger real material datasets
- integrating neural intrinsic decomposition networks

- extending the system to support additional materials such as glass or fabric
- incorporating neural radiance fields for multi-view material editing
- improving segmentation accuracy using advanced models such as SAM2
- introducing quantitative evaluation metrics for material realism.

13 Conclusion

This project presented a hybrid pipeline for prompt-guided material editing of objects in images. Initial experiments using diffusion models as the primary editing mechanism revealed fundamental limitations related to geometry preservation and material representation.

The final system combines object segmentation, intrinsic image decomposition, frequency-domain material manipulation, and optional diffusion refinement. This hybrid approach enables realistic material transformations while preserving object geometry and scene context.

The results demonstrate that integrating physically meaningful image representations with generative refinement provides a promising direction for structured image editing tasks.

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